Day 3: DNS Server ConfigurationTheory (30 min)

1.Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Aim

To install and configure a local DNS server using BIND on Linux or Simple DNS Plus on Windows and create a new DNS zone named musicmania.in.

Theory

DNS (Domain Name System) translates human-readable domain names, such as musicmania.in, into IP addresses that computers use to communicate over a network.

BIND (Berkeley Internet Name Domain) is DNS server software commonly used on Linux. A DNS zone is a portion of the domain namespace for which a DNS server stores and manages DNS records.

Procedure A: Using BIND on Linux (Ubuntu)

Step 1: Install BIND

Open the terminal and execute:

sudo apt update

sudo apt install bind9 bind9-utils dnsutils

Step 2: Create a DNS zone

Open the local DNS configuration file:

sudo nano /etc/bind/named.conf.local

Add the following configuration:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Save the file.

Step 3: Create the zone file

sudo nano /etc/bind/db.musicmania.in

Add these sample DNS records:

```
$TTL 86400
@   IN  SOA ns1.musicmania.in. admin.musicmania.in. (
        2026100901
        3600
        1800
        604800
        86400
)
@       IN  NS      ns1.musicmania.in.
ns1     IN  A       192.168.1.10
@       IN  A       192.168.1.10
www     IN  A       192.168.1.10
```

Here, 192.168.1.10 is an example IP address. Replace it with the actual IP address of your DNS server or the local web server you want to represent.

Step 4: Validate the configuration

```
sudo named-checkconf
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

If the zone file is valid, the zone check should report OK.

Step 5: Restart the DNS service

```
sudo systemctl restart bind9
sudo systemctl status bind9
```

Step 6: Test DNS resolution

```
dig @127.0.0.1 musicmania.in
dig @127.0.0.1 www.musicmania.in
```

The DNS response should contain the sample IP address 192.168.1.10.

Procedure B: Using Simple DNS Plus on Windows

1. Download and install Simple DNS Plus on your Windows computer.
2. Open the DNS server management interface.
3. Create a new primary zone named musicmania.in.
4. Add an A record for musicmania.in pointing to a sample IP address, such as 192.168.1.10.
5. Add an A record for www.musicmania.in pointing to the same IP address.
6. Add an NS record for ns1.musicmania.in and the corresponding A record for the name server.
7. Save the configuration and ensure the DNS service is running.
8. Test the domain using nslookup, specifying the DNS server's IP address.

Example:

nslookup musicmania.in 127.0.0.1

nslookup www.musicmania.in 127.0.0.1

Note: These are local demonstration records. Creating a local zone does not automatically make musicmania.in accessible publicly on the internet. For public access, domain registration, delegation, and appropriate DNS/network configuration are also required.

Sample observations

Parameter	**Observation**
DNS software	BIND / Simple DNS Plus
Domain name	musicmania.in
Zone type	Primary (master) zone
Name server	ns1.musicmania.in
Sample IP address	192.168.1.10
DNS record types	SOA, NS, A
Expected result	Domain resolves to the configured IP address

2.Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works

Aim

To add an A record in the DNS server configuration that maps www.musicmania.in to the IP address 192.168.1.50 and verify it using nslookup or dig.

Theory

An A (Address) record maps a domain name to an IPv4 address. It allows users to access a server using a domain name instead of entering its IP address.

Procedure (BIND on Linux)

Step 1: Open the DNS zone file

sudo nano /etc/bind/db.musicmania.in

Step 2: Add the A record

Add the following line to the zone file:

www IN A 192.168.1.50

Step 3: Update the serial number

In the SOA record, increment the serial number, for example:

2026100902

Step 4: Check and restart the DNS server

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

sudo systemctl restart bind9

Step 5: Verify using nslookup

Run the following command, replacing 127.0.0.1 if your DNS server uses a different IP address:

nslookup www.musicmania.in 127.0.0.1

Expected output:

Server: 127.0.0.1

Address: 127.0.0.1#53

Name: www.musicmania.in

Address: 192.168.1.50

Alternatively, use dig:

dig @127.0.0.1 www.musicmania.in A

The answer section should contain:

www.musicmania.in. 86400 IN A 192.168.1.50

Observation

Parameter	**Value**
Domain name	www.musicmania.in
Record type	A
Configured IP address	192.168.1.50
Verification command	nslookup / dig
Expected result	Domain resolves to 192.168.1.50

3.Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Aim

To add a CNAME record for play.musicmania.in pointing to www.musicmania.in and verify whether it resolves to the correct IP address.

Theory

A CNAME (Canonical Name) record is used to create an alias for another domain name. In this practical, play.musicmania.in will act as an alias for www.musicmania.in, which points to the IP address 192.168.1.50.

Procedure (BIND on Linux)

Step 1: Open the DNS zone file

sudo nano /etc/bind/db.musicmania.in

Step 2: Add the CNAME record

Add the following line to the zone file:

play IN CNAME www.musicmania.in.

Ensure that www.musicmania.in already has an A record pointing to 192.168.1.50. Do not create another A record for play because a CNAME cannot coexist with other ordinary records at the same name.

Step 3: Update the serial number

Increase the SOA serial number, for example, from 2026100902 to 2026100903.

Step 4: Validate and restart the DNS server

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

sudo systemctl restart bind9

Step 5: Test using nslookup

nslookup play.musicmania.in 127.0.0.1

Expected output (format may vary):

Server: 127.0.0.1

Address: 127.0.0.1#53

Name: www.musicmania.in

Address: 192.168.1.50

Aliases: play.musicmania.in

Alternatively, use:

dig @127.0.0.1 play.musicmania.in

The answer section should show the CNAME pointing to www.musicmania.in and its A record resolving to 192.168.1.50.

Observation

Parameter	**Value**
Alias domain	play.musicmania.in
Record type	CNAME
Target domain	www.musicmania.in
Target IP address	192.168.1.50

Verification tool	nslookup / dig
Expected result	Alias resolves to 192.168.1.50

**4.Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.**

Aim

To troubleshoot DNS resolution problems for musicmania.in by checking DNS configuration files, verifying the DNS service, and testing domain resolution using nslookup and dig.

Theory

DNS (Domain Name System) translates domain names into IP addresses. If musicmania.in does not resolve correctly, the problem may be caused by incorrect zone records, syntax errors, a stopped DNS service, or an incorrect DNS server address.

Step-by-Step Troubleshooting Procedure

Step 1: Check the DNS configuration file

Open the BIND local configuration file:

sudo nano /etc/bind/named.conf.local

Verify that the zone is configured correctly:

```
zone "musicmania.in" {
   type master;
   file "/etc/bind/db.musicmania.in";
};
```

Observation: Confirm that the domain name and zone file path are correct.

Step 2: Check the zone file

Run:

sudo nano /etc/bind/db.musicmania.in

Verify that the zone contains the correct records:

@ IN NS ns1.musicmania.in.

ns1 IN A 192.168.1.10

@ IN A 192.168.1.50

www IN A 192.168.1.50

Observation: Check that the required A and NS records are present and that IP addresses are correct for your intended setup.

Step 3: Validate the DNS configuration

Execute:

sudo named-checkconf

Then validate the zone file:

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

Observation: The configuration check should return no errors, and the zone check should report OK. If errors appear, correct the syntax or record format.

Step 4: Check the DNS service status

Run:

sudo systemctl status bind9

Observation: Check whether the BIND service is active and running. If the service is stopped or failed, investigate the reported error.

Step 5: Restart the DNS service

After correcting any configuration errors, run:

sudo systemctl restart bind9

Verify the service again:

sudo systemctl status bind9

Observation: Confirm that the DNS service has restarted successfully.

Step 6: Test domain resolution using nslookup

Execute:

nslookup musicmania.in 127.0.0.1

Test the www subdomain as well:

nslookup www.musicmania.in 127.0.0.1

Expected result: The domain should resolve to the IP address configured in the zone file.

Step 7: Test using dig

Run:

dig @127.0.0.1 musicmania.in A

Check the ANSWER SECTION in the output. It should display the configured A record and IP address.

If the answer is missing or incorrect, inspect the zone records and verify that you are querying the correct DNS server.

Step 8: Check DNS logs if the problem persists

Run:

sudo journalctl -u bind9 --no-pager -n 30

Observation: Look for errors related to zone loading, syntax, permissions, or port conflicts.

Troubleshooting Summary

Problem	**Troubleshooting action**
Incorrect domain configuration	Check /etc/bind/named.conf.local
Invalid zone file	Run named-checkzone
DNS service stopped	Check status and restart BIND
Incorrect IP address returned	Verify A records and zone serial number
DNS query fails	Test with nslookup and dig
Configuration errors persist	Review BIND service logs